

M359 University relational representation



The Open University

This document contains the full relational representation for the University model. This representation was developed by following the techniques taught in *Block 2* of the course material.

The document is supplied so that you can see a complete relational declaration for the University scenario and related conceptual data model. A summary of the relational headings for the University representation can be found on the *Relational headings summary* card, and the relational extension that completes this representation can also be found on the *University data summary* card.

This document should be read in conjunction with the course material in *Block 2*.

University relational declaration

Model *University*

Domains

RegionNumbers = { 1... 12}
Addresses
TelephoneNumbers = {strings of 15 numerals}
EmailAddresses = {string@string}
StudentIds = {s01...s99}
Names = {Family Name}
TitlesOfCourses = {string of alphabetic characters}
Dates = standard dates
StaffNumbers = {string of 4 numerals}
CourseCodes = {c1...c9}
Credits = {30, 60}
{c.5 The attribute *Credit*, of *Course*, has a value of 30 or 60 points.}
Limits = Integer
AssignmentNumbers = {1, 2, 3, 4, 5}
Percentages = {0... 100}
{c.7 The *Mark* attributes (of *Assignment* and *Examination*) have values that represent marks between 0 and 100.}
Locations = {string of alphabetic characters}
FormNumbers = {SCxyz, where x, y, z are numerals}

We defer until implementation the consideration of whether we wish an address to be a single data object or a more complex object.

relation *Region*

RegionNumber: *RegionNumbers*
Address: *Addresses*
Telephone: *TelephoneNumbers*
EmailAddress: *EmailAddresses*
primary key *RegionNumber*

relation *Staff*

StaffNumber: *StaffNumbers*
Name: *Names*
Address: *Addresses*
EmailAddress: *EmailAddresses*
Telephone: *TelephoneNumbers*
RegionNumber: *RegionNumbers*
primary key *StaffNumber*
{mandatory participation of *Staff* in *ContractsWith* relationship}
foreign key *RegionNumber* **references** *Region*

relation *TelephoneTutoredBy* {relation represents *TelephoneTutoredBy* relationship}

StaffNumber: *StaffNumbers*
CourseCode: *CourseCodes*
primary key *StaffNumber*
foreign key *StaffNumber* **references** *Staff*
foreign key *CourseCode* **references** *Course*

relation *Course*

CourseCode: *CourseCodes*
Title: *TitlesOfCourses*
Credit: *Credits*
primary key *CourseCode*
{mandatory participation of *Course* in *TelephoneTutoredBy* relationship}
constraint (
 (**project** *Course* **over** *CourseCode*)
 difference
 (**project** *TelephoneTutoredBy* **over** *CourseCode*)
)**is empty**

relation *Quota*

CourseCode: CourseCodes

Limit: Limits

DateReviewed: Dates

primary key *CourseCode*

{mandatory participation of Quota in LimitedBy relationship}

foreign key *CourseCode* **references** *Course*

relation *Enrolment*

StudentId: StudentIds

CourseCode: CourseCodes

EnrolmentDate: Dates

primary key *(StudentId, CourseCode)*

{mandatory participation of Enrolment in StudiedBy relationship}

foreign key *CourseCode* **references** *Course*

{mandatory participation of Enrolment in EnrolledIn relationship}

foreign key *StudentId* **references** *Student*

relation *Tutors {relation represents Tutors relationship}*

StudentId: StudentIds

CourseCode: CourseCodes

StaffNumber: StaffNumbers

primary key *(StudentId, CourseCode)*

foreign key *StaffNumber* **references** *Staff*

foreign key *StudentId, CourseCode* **references** *Enrolment*

relation *Student*

StudentId: StudentIds

Name: Names

Address: Addresses

EmailAddress: EmailAddresses

RegistrationDate: Dates

RegionNumber: RegionNumbers

primary key *StudentId*

{mandatory participation of Student in Manages relationship}

foreign key *RegionNumber* **references** *Region*

relation *Assignment*

StudentId: StudentIds

CourseCode: CourseCodes

AssignmentNumber: AssignmentNumbers

Mark: Percentages

primary key *(StudentId, CourseCode, AssignmentNumber)*

{mandatory participation of Assignment in Submits relationship}

foreign key *StudentId, CourseCode* **references** *Enrolment*

relation *Examination*

StudentId: StudentIds

CourseCode: CourseCodes

ExaminationLocation: Locations

Mark: Percentages

primary key *StudentId, CourseCode*

{mandatory participation of Examination in Takes relationship}

foreign key *StudentId, CourseCode* **references** *Enrolment*

relation *SpecialCircumstancesForm*

FormNumber: FormNumbers

DateReceived: Dates

primary key *FormNumber*

relation *AssignmentIncludes* {relation represents AssignmentIncludes relationship}
StudentId: StudentIds
CourseCode: CourseCodes
AssignmentNumber: AssignmentNumbers
FormNumber: FormNumbers
primary key *StudentId*, *CourseCode*, *AssignmentNumber*
 {relationship AssignmentIncludes is 1:1}
alternate key *FormNumber*
foreign key *StudentId*, *CourseCode*, *AssignmentNumber* **references** *Assignment*
foreign key *FormNumber* **references** *SpecialCircumstancesForm*

relation *ExamIncludes* {relation represents ExamIncludes relationship}
StudentId: StudentIds
CourseCode: CourseCodes
FormNumber: FormNumbers
primary key *StudentId*, *CourseCode*
 {relationship ExamIncludes is 1:1}
alternate key *FormNumber*
foreign key *StudentId*, *CourseCode* **references** *Examination*
foreign key *FormNumber* **references** *SpecialCircumstancesForm*

{c.1 A staff member may only tutor students who are located in the region for which the member of staff has a contract.}

constraint (
select
 (*Tutor*
join
 (**project** *Student* **over** *StudentId*, *RegionNumber*
rename (*RegionNumber* **as** *Region*))
join
Staff)
where *Region* <> *RegionNumber*
) is empty

{c.2 A staff member who acts as a telephone tutor must also tutor at least one enrolment on the same course.}

constraint (
 (**project** *TelephoneTutoredBy* **over** *StaffNo*, *CourseCode*)
difference
 (**project** *Tutors* **over** *StaffNo*, *CourseCode*)
) is empty

{c.4 Students may not enrol for more than 180 points' worth of courses at any one time.}

{Note: This constraint cannot be represented in the relational representation as it requires arithmetic operators (to sum the number of credit points for a course) and these are not available in relational algebra.}

{c.3 A student's RegistrationDate must be before the same student's EnrolmentDate for any Enrolment.}

constraint (
select
 (*Enrolment* **join** *Student*)
where *RegistrationDate* >= *EnrolmentDate*
) is empty

{c.6 A 60-point course may have up to five assignments and a 30-point course up to three assignments.}

```

constraint (
  select
    (Assignment join Course)
  where ((Credit = 60 and AssignmentNumber > 5)
    or (Credit = 30 and AssignmentNumber > 3))
) is empty

```

{c.8 A FormNumber cannot be involved in both the AssignmentIncludes and the ExamIncludes relationships, but must be involved in one of these relationships.}

```

constraint (
  ( (project AssignmentIncludes over FormNumber)
    intersection
    (project ExamIncludes over FormNumber))
  union
  ( (project SpecialCircumstancesForm over FormNumber)
    difference
    ((project AssignmentIncludes over FormNumber)
      union
      (project ExamIncludes over FormNumber)))
) is empty

```

{c.9 DateReceived on SpecialCircumstancesForm must be on or before today's date.}

```

constraint (
  select SpecialCircumstancesForm
  where DateReceived > $Today'sDate$
) is empty

```

{c.10 DateReviewed for Quota must be on or before today's date.}

```

constraint (
  select Quota
  where DateReviewed > $Today'sDate$
) is empty

```

{Constraints c.11, c.12, c.13, c.14 and c.15 are represented by the foreign key and primary key declarations.}

{Note: In the above, \$Today'sDate\$ represents the value of the current date.}

The University relational headings

Region (RegionNumber, Address, Telephone, EmailAddress)

Staff (StaffNumber, Name, Address, EmailAddress, Telephone, RegionNumber)

TelephoneTutoredBy (StaffNumber, CourseCode)

Course (CourseCode, Title, Credit)

Quota (CourseCode, Limit, DateReviewed)

Enrolment (StudentId, CourseCode, EnrolmentDate)

Tutors (StudentId, CourseCode, StaffNumber)

Student (StudentId, Name, Address, EmailAddress, RegistrationDate, RegionNumber)

Assignment (StudentId, CourseCode, AssignmentNumber, Mark)

Examination (StudentId, CourseCode, ExaminationLocation, Mark)

SpecialCircumstancesForm (FormNumber, DateReceived)

AssignmentIncludes (StudentId, CourseCode, AssignmentNumber, FormNumber)

ExamIncludes (StudentId, CourseCode, FormNumber)

University relational extension

Region

RegionNumber	Address	Telephone	EmailAddress
1	57, Longboat Street, Birmingham	0120 779165	region1@open.fake.address
2	Sumner House, Curtis Street, Manchester	0160 91659	region2@open.fake.address
3	Block 9, The Campus, Walton Hill,	01670 245365	region3@open.fake.address
4	Suite 2, Faculty Towers, Torquay	02563 13829	region4@open.fake.address
12	The Office, New York	10898 227191	region12@open.fake.address

Staff

StaffNumber	Name	Address	EmailAddress	Telephone	RegionNumber
8431	French	123 Townstreet, Roadville	french@open.tutors.fake	09273 28291	4
7774	Redhead	17 Boot Lane, Wellesley	redhead@myownemail.com	01281 19323	4
5212	Heathcote	234 Rue de stad, Alphaville	heathcote2@herelam.co.uk	08104 31221	4
3158	Jennings	21 Beckett Way, Mitreton	artoo@mechadroid.net	04323 342322	3
5324	Lai	53 Broadwalk, New York	lai2@well.parc.co.uk	03223 43652	4
3678	Sanderson	56 Railway Cuttings, Pottsford	helen@open.tutors.fake	08642 92753	1
4219	Sanderson	23 Bridge Road, Hazeltown	sandy@basketball.fake.co.uk	063245 93725	2
6451	Clark	16 Beasley Street, Chickentown	jcc@salford.fake.gov.uk	0826 82074	1

TelephoneTutoredBy

<i>StaffNumber</i>	<i>CourseCode</i>
7774	c5
8431	c2
6451	c2
3158	c4
5324	c7

Course

<i>CourseCode</i>	<i>Title</i>	<i>Credit</i>
c2	Syntax	30
c4	Semantics	60
c7	Pragmatics	30
c5	Logic	60

Quota

<i>CourseCode</i>	<i>Limit</i>	<i>DateReviewed</i>
c2	3000	10/01/2007
c4	250	10/01/2007
c5	250	23/04/2006

Enrolment

<i>StudentId</i>	<i>CourseCode</i>	<i>EnrolmentDate</i>
s01	c4	Jan 12, 2005
s02	c5	Jan 1, 2005
s02	c7	Jun 12, 2005
s05	c2	Jun 4, 2004
s05	c7	Oct 18, 2004
s07	c4	Dec 12, 2004
s09	c4	Dec 16, 2004
s09	c2	Dec 18, 2004
s09	c7	Dec 15, 2004
s10	c7	Jun 20, 2004
s10	c4	May 5, 2004
s22	c2	Mar 15, 2002
s38	c2	Sep 18, 2003
s38	c5	Mar 9, 2004
s46	c2	Mar 1, 2002
s57	c4	Jun 30, 2001
s57	c5	Jan 20, 2003

Tutors

<i>StudentId</i>	<i>CourseCode</i>	<i>StaffNumber</i>
s01	c4	3158
s02	c5	5324
s02	c7	5212
s05	c2	5324
s05	c7	5324
s07	c4	3158
s09	c4	5324
s09	c2	8431
s09	c7	5324
s10	c7	5324
s10	c4	5324
s22	c2	6451
s38	c2	3678
s38	c5	3678
s46	c2	4219
s57	c4	5212
s57	c5	7774

Student

<i>StudentId</i>	<i>Name</i>	<i>Address</i>	<i>EmailAddress</i>	<i>RegistrationDate</i>	<i>RegionNumber</i>
s01	Akeroyd	12 Anystreet, Anytown	Akers@tahoo.fake.com	Nov 23, 1999	3
s02	Thompson	8 High Street, Lowville	Pjay@thompson.fake.com	Oct 12, 2004	4
s05	Ellis	34 Globe Road, Smallville	G.Ellis@fake.fake.co.uk	Oct 14, 2003	4
s07	Gillies	29 Straight Street, Angletown	Gillies@address.fake.net	Dec 2, 1993	3
s09	Reeves	34 The Crescent, Curville	T.Reeves@nnet.fake.com	Dec 14, 2004	4
s10	Urbach	22 Hilltops, Valley Town	B.Urbach@tnet.fake.fr	May 5, 2003	4
s22	Bryant	84 Brook Street, Little Hacking	A.Bryant@greenmail.fake.uk	Jun 21, 2000	1
s38	Patel	12 Stanley Road, Pitchford	patel122@mailman.fake.uk	Oct 8, 2001	1
s42	Reddick	23 Kestrel Lane, Dudley	dave@belwise.fake.co.uk	Apr 23, 2002	2
s46	Sharp	The Farm, Lower Watley	F.Sharp@bluesky.fake.ac.uk	Feb 14, 2002	2
s57	Patel	4 Lower Crescent, Cinderfield	r.patel@tahoo.fake.com	Nov 5, 2000	4

Assignment

<i>StudentId</i>	<i>CourseCode</i>	<i>AssignmentNumber</i>	<i>Mark</i>
<i>s01</i>	<i>c4</i>	<i>1</i>	<i>80</i>
<i>s01</i>	<i>c4</i>	<i>2</i>	<i>74</i>
<i>s02</i>	<i>c5</i>	<i>1</i>	<i>70</i>
<i>s02</i>	<i>c5</i>	<i>2</i>	<i>65</i>
<i>s02</i>	<i>c5</i>	<i>3</i>	<i>73</i>
<i>s02</i>	<i>c7</i>	<i>1</i>	<i>53</i>
<i>s02</i>	<i>c7</i>	<i>2</i>	<i>58</i>
<i>s05</i>	<i>c2</i>	<i>1</i>	<i>78</i>
<i>s05</i>	<i>c2</i>	<i>2</i>	<i>63</i>
<i>s07</i>	<i>c4</i>	<i>1</i>	<i>91</i>
<i>s09</i>	<i>c4</i>	<i>1</i>	<i>85</i>
<i>s09</i>	<i>c2</i>	<i>1</i>	<i>92</i>
<i>s09</i>	<i>c2</i>	<i>2</i>	<i>76</i>
<i>s10</i>	<i>c4</i>	<i>1</i>	<i>45</i>
<i>s22</i>	<i>c2</i>	<i>1</i>	<i>74</i>
<i>s22</i>	<i>c2</i>	<i>2</i>	<i>68</i>
<i>s38</i>	<i>c5</i>	<i>1</i>	<i>51</i>
<i>s38</i>	<i>c5</i>	<i>3</i>	<i>43</i>
<i>s57</i>	<i>c4</i>	<i>1</i>	<i>93</i>
<i>s57</i>	<i>c4</i>	<i>2</i>	<i>82</i>
<i>s57</i>	<i>c4</i>	<i>3</i>	<i>84</i>
<i>s57</i>	<i>c5</i>	<i>1</i>	<i>74</i>

Examination

<i>StudentId</i>	<i>CourseCode</i>	<i>ExaminationLocation</i>	<i>Mark</i>
<i>s07</i>	<i>c4</i>	<i>Bedford</i>	<i>85</i>
<i>s09</i>	<i>c4</i>	<i>Taunton</i>	<i>63</i>
<i>s10</i>	<i>c4</i>	<i>Gateshead</i>	<i>27</i>
<i>s09</i>	<i>c7</i>	<i>Taunton</i>	<i>71</i>
<i>s05</i>	<i>c2</i>	<i>Bath</i>	<i>57</i>
<i>s09</i>	<i>c2</i>	<i>New York</i>	<i>56</i>

SpecialCircumstancesForm

<i>FormNumber</i>	<i>DateReceived</i>
<i>SC458</i>	<i>16-04-2004</i>
<i>SC754</i>	<i>07-03-2004</i>
<i>SC245</i>	<i>19-10-2004</i>

AssignmentIncludes

<i>StudentId</i>	<i>CourseCode</i>	<i>AssignmentNumber</i>	<i>FormNumber</i>
<i>s09</i>	<i>c2</i>	<i>1</i>	<i>SC754</i>
<i>s10</i>	<i>c4</i>	<i>1</i>	<i>SC458</i>

ExamIncludes

<i>StudentId</i>	<i>CourseCode</i>	<i>FormNumber</i>
<i>S05</i>	<i>c2</i>	<i>SC245</i>